

Assignment 6.2

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Batch – 03

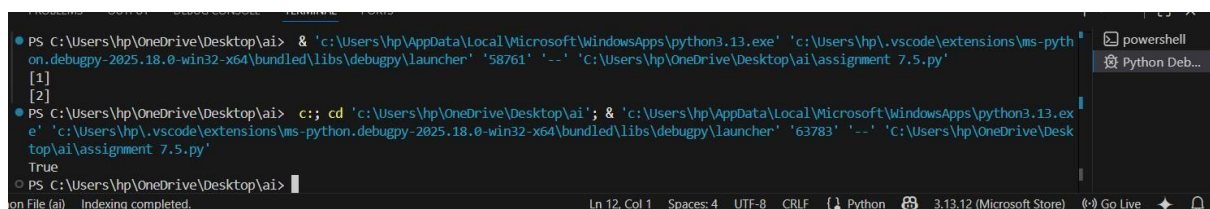
Task 1: (Classes – Data Validation)

- Prompt AI to generate a Student class with attributes: name, roll_no, and marks. Add a method is_pass() that returns whether the student has passed (marks $\geq$ 40).

```
Fact.py > ...
1 class Student:
2     def __init__(self, name, roll_no, marks):
3         self.name = name
4         self.roll_no = roll_no
5         self.marks = marks
6
7     def is_pass(self):
8         return self.marks >= 40
9
10
11 s1 = Student("Akshith", 1177, 65)
12 print(s1.is_pass())
```

Expected Output-1

- Python class with constructor and validation logic
- Clear method implementation returning pass/fail status
- Students analyze correctness and clarity of class design



```
PS C:\Users\hp\OneDrive\Desktop\ai> & 'c:\Users\hp\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\hp\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58761' '--' 'c:\Users\hp\OneDrive\Desktop\ai\assignment 7.5.py'
[1]
[2]
PS C:\Users\hp\OneDrive\Desktop\ai> c:: cd 'c:\Users\hp\OneDrive\Desktop\ai'; & 'c:\Users\hp\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\hp\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '63783' '--' 'c:\Users\hp\OneDrive\Desktop\ai\assignment 7.5.py'
True
PS C:\Users\hp\OneDrive\Desktop\ai>
```

Task 2: (Loops – Pattern Generation)

Ask AI to generate a function that prints a right-angled triangle star pattern using a for loop. Then regenerate the same pattern using a while loop.

```
assignment 7.5.py > ...
1  def triangle_while(n):
2      i = 1
3      while i <= n:
4          print(" * " * i)
5          i += 1
6
7  triangle_while(5)
8  # Task 5: Privacy-Aware Data Logging
```

Output-2

- Correct pattern output using both loop types
- Logical loop structure with proper conditions

```
PS C:\Users\hp\OneDrive\Desktop\ai> c::; cd 'c:\Users\hp\OneDrive\Desktop\ai'; & 'c:\Users\hp\AppData\Local\Microsoft\Windows\Terminal\1000\powershell.exe' 'c:\Users\hp\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '49562' 'c:\Users\hp\OneDrive\Desktop\ai\assignment 7.5.py'
*
**
***
****
*****
PS C:\Users\hp\OneDrive\Desktop\ai> 
```

Task 3: (Conditional Statements – Number Analysis)

Ask AI to write a function that checks whether a given number is positive, negative, or zero using if-elif-else. Test the function with multiple inputs.

```
assignment 7.5.py > ...
1  def check_number(num):
2      if num > 0:
3          return "Positive"
4      elif num < 0:
5          return "Negative"
6      else:
7          return "Zero"
8
9  print(check_number(10))
10 print(check_number(-5))
11 print(check_number(0))
12
```

Output-3

- Function correctly classifies numbers
- Proper handling of all conditions
- Students analyze decision logic

```
PS C:\Users\hp\OneDrive\Desktop\ai> c::; cd 'c:\Users\hp\OneDrive\Desktop\ai'; & 'c:\Users\hp\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.11_x-ww\python.exe' 'c:\Users\hp\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '5394' 'c:\Users\hp\OneDrive\Desktop\ai\assignment 7.5.py'
Positive
Negative
Zero
PS C:\Users\hp\OneDrive\Desktop\ai>
```

Task 4: (Nested Conditionals)

- Generate a function `check_discount(age, is_member)` that determines discount eligibility:
- Age $\geq 60 \rightarrow$ Senior discount
 - Member $\rightarrow$ Additional discount Use nested if statements.

```
Welcome | day 1.py M | assignment 5.5.py U | assignment 7.5.py U x | task1.py M | assignment.py U | day 1.py (Index) | D | | | ...
assignment 7.5.py > ...
1 def check_discount(age, is_member):
2     if age >= 60:
3         if is_member:
4             return "Senior + Member Discount"
5         else:
6             return "Senior Discount"
7     else:
8         if is_member:
9             return "Member Discount"
10        else:
11            return "No Discount"
12
13 print(check_discount(65, True))
14 print(check_discount(30, False))
15 # Task 5: Privacy-Aware Data Logging
```

Output-4

- Python code using nested conditionals
- Clear explanation of decision flow

```
PS C:\Users\hp\OneDrive\Desktop\ai> c::; cd 'c:\Users\hp\OneDrive\Desktop\ai'; & 'c:\Users\hp\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\hp\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '56979' '--' 'c:\Users\hp\OneDrive\Desktop\ai\assignment 7.5.py'
Senior + Member Discount
No Discount
PS C:\Users\hp\OneDrive\Desktop\ai> |
```

Task 5: (Class – Mathematical Opera)

- Ask AI to create a Circle class with methods to calculate area () and circumference () given the radius.

```
assignment 7.5.py > ...
1 import math
2
3 class Circle:
4     def __init__(self, radius):
5         self.radius = radius
6
7     def area(self):
8         return math.pi * self.radius * self.radius
9
10    def circumference(self):
11        return 2 * math.pi * self.radius
12
13 c = Circle(7)
14 print(c.area())
15 print(c.circumference())
16 # Task 5: Privacy-Aware Data Logging
```

Output-5

- Correct mathematical computation

- Well-structured class with methods
- Code explanation provided

```
PS C:\Users\hp\OneDrive\Desktop\ai> c::; cd 'c:\Users\hp\OneDrive\Desktop\ai'; & 'c:\Users\hp\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\hp\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '59204' '--' 'C:\Users\hp\OneDrive\Desktop\ai\assignment_7.5.py'
153.93804002589985
43.982297150257104
PS C:\Users\hp\OneDrive\Desktop\ai>
```

Ln 16, Col 1 Spaces: 4 UTF-8 CRLF Python 3.13.12 (Microsoft Store) Go Live